

Multi-perspective learning for enhanced user preferences for session-based recommendation

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ABSTRACT

Due to the sparsity and anonymity of interactions within sessions, capturing user preferences by modeling item relationships poses a significant challenge. Previous methods have attempted to utilize collaborative information from adjacent sessions or item attribute knowledge as supplements. However, these methods are prone to introducing interfering information through their session similarity measurement strategies and attribute embedding learning processes, both requiring optimization. Moreover, they often fail to fully model global item relationships. To address these issues, we propose a novel method called Multi-perspective Learning for Enhanced User Preferences (MLEUP) for session-based recommendations. Specifically, we first construct reliable session groups by measuring the overlap and incompatibility of items across sessions to calculate session similarity. Then, by modeling attribute-item and attribute-attribute relationships, along with employing a neighbor-based contrastive self-supervised attribute learning task, we learn high-quality attribute embeddings that capture the user's stable preferences. Next, leveraging the session groups, we model global item multi-granularity co-occurrence and high-order sequential relationships through designed attention layers, enabling us to learn informative item embeddings that represent the user's dynamic preferences. Finally, we further model the correlated relationship between the two user preference representations. Our approach ensures that effective attribute knowledge and global collaborative information are learned to mitigate the data sparsity issue, while comprehensively modeling global item relationships to capture the user's complex dynamic preferences. Experimental results demonstrate that MLEUP achieves state-of-the-art performance in terms of precision, MRR, and NDCG. The experimental datasets and source code of MLEUP are publicly available at <https://github.com/lean030/MLEUP>.

1. Introduction

Traditional recommendation systems are effective tools for alleviating the problem of information overload, and are dedicated to solving the problem of recommending static user preferences by relying on user profiles to provide personalized services. They are widely used in many fields. However, in the real world of online services, on the one hand, due to privacy regulations, more and more user identities are not disclosed, and on the other hand, user preferences for items are often dynamic and change over time. Therefore, session-based recommendation systems (SBRs) have emerged. They are applied in anonymous, short interaction scenarios, without relying on user profiles, to capture user preferences contained in the anonymous session and recommend the next item to the user, facing two challenges.

The first challenge is data sparsity. Due to the short duration of sessions and the large number of candidate items typically available on the platform, the interactions within a single session are often very

limited, and many of them are long-tail items with very few total interaction counts. For example, in the commonly used Tmall dataset, the average session length is less than 7, and there are 40,727 candidate items, with over 80% of items having only around 10 interactions. There are two approaches to addressing the problem of data sparsity:

- The first approach is to supplement the given session with collaborative information from other sessions [1–5]. One way to do this is by finding neighboring items on a global scale, and another way is to utilize similarity measurement methods such as radial basis function kernels, cosine similarity, and item repetition across sessions to find neighboring sessions for a given session, then study the propagation of information at the session level or mine the supervised information provided by the sessions. However, these methods can encounter issues with collecting interfering information. If two sessions reflect significantly different intents, for

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example, one session “flour”, “flowers”, “sugar”, “candles” where the user may want to make a flower cake, and the current session “balloons”, “flowers”, “candles” where the user may want to purchase decorations for an event, collecting “flour” and “sugar” as global neighboring items for “flowers” would clearly contradict the intent of using flowers as decoration in the current session. Therefore, it is crucial to finely measure the similarity between sessions and find more reliable neighbors for each session in order to effectively mine collaborative information based on that.

- The second approach is to supplement additional knowledge by incorporating attributes of the items (such as category, price and description text for a product, or artist for a song). The research on integrating attribute knowledge in SBRs has just begun [6–9]. The attributes of the items often reflect the general preferences of the users. For example, users may have a preference for a specific category, brand, or price range of products, or they may prefer songs by a particular artist. Therefore, we learn attribute knowledge. On the one hand, we generate session attribute embedding as stable user preference. On the other hand, we integrate attribute embeddings into the session relational embeddings learning process to guide the generation of two types of session relational embeddings, thereby influencing dynamic user preference. In this case, the learning of attribute embeddings and item embeddings is tightly coupled, and the quality of attribute embeddings significantly affects the generation of accurate user preferences and thus the performance of recommendations. Therefore, we consider constructing a neighbor-based contrastive self-supervised attribute learning task to explore additional supervised signals among sessions to improve the quality of attribute embeddings.

The second challenge is that the user interests revealed by sessions are often dynamic and complex, requiring modeling of the relationships between items. First of all, items within the same session have a co-occurrence relationship, and sequences of different granularity may reveal different user preferences. For example, given a session of “flour”, “sugar”, “flowers”, if only a single item in the session, such as “flowers”, is considered, the user may want to purchase some decoration, and a reasonable recommendation for the next item would be balloons. When considering a sequence of length 2, such as “sugar” and “flowers”, the user may want to make jam, and a reasonable recommendation for the next item would be a non-stick pan. When considering a sequence of length 3, the user may want to make flower cakes, and a reasonable recommendation for the next item would be an oven. Therefore, modeling the co-occurrence relationship of items at multiple granularities is effective for capturing the complex intentions of users. In addition, the transition relationship between items may have a strict order, for example, given a session of “TV”, “TV stand”, “speakers”, users usually purchase a “TV stand” after purchasing a “TV”, and after purchasing the first two, they will then buy a “speaker” that matches them. This indicates that a click on one item may depend on the transitions of multiple items. Therefore, modeling a high-order sequential relationship between items is necessary. In this paper, we construct a session hypergraph and a session heterogeneous graph to capture two types of relationships between items. By combining the results of attribute embedding learning, we generate two types of session relationship embeddings, thereby obtaining the user’s dynamic preference.

In summary, to tackle these two challenges, we propose a novel method, namely multi-perspective learning for enhanced user preferences (MLEUP) that mitigates the problem of data sparsity by incorporating global collaborative information and attribute knowledge, to generate the user’s stable preference. Moreover, we capture the user’s dynamic preference by comprehensively modeling multi-granularity item co-occurrence relationship and high-order item sequential relationship. To sum up, our contributions in this paper are as follows:

- We propose a session similarity measurement strategy that combines positive and negative relationships, namely the implicit cross-session item incompatibility relationship and the explicit cross-session item overlap relationship. Therefore, this strategy ensures the integration of collaborative information across sessions while avoiding excessive interference.
- We propose an effective attribute embeddings learning model that captures global attribute–item relationship and intra-session attribute–attribute relationship. Importantly, we construct a novel neighbor-based contrastive self-supervised attribute learning task that enhances the quality of attribute embeddings and generates more accurate user preferences, thereby improving recommendation performance.
- We propose a comprehensive item embedding learning model that captures global multi-granularity item co-occurrence relationship and global high-order item sequential relationship. Notably, we innovatively introduce relative position encoding technique that capture more accurate and complete high-order sequential relationship among items, aiding in learning the information propagated through this process.
- Extensive experiments are conducted on three benchmark datasets, i.e., Tmall, Diginetica and 30music. The experimental results demonstrate the superiority of MLEUP over the state-of-the-art baselines in terms of precision, MRR and NDCG. The experimental datasets and the source code of MLEUP are shared on GitHub.¹

2. Related work

In this section, we review the related work for session-based recommendations (SBRs).

Traditional session-based recommendation (SBR) methods. Traditional SBR methods include Markov Chain-based methods [10,11] and machine learning methods such as the nearest neighbor model [12] and the linear model [13]. Rendle et al. [10] proposed combining matrix factorization and Markov chain to learn a personalized transition matrix for each user and achieve personalized recommendations, but it only applies to non-anonymous data. Kersbergen et al. [12] proposed the VMIS-kNN model, which is an improvement on the VS-KNN model. It uses indexing and two technical methods (minimizing intermediate results and early stopping) to efficiently process large volumes of recommendation requests. Choi et al. [13] proposed to utilize the closed-form solution of linear models for recommendation, which has high scalability. However, traditional methods cannot flexibly and effectively assign weights to items in the session, and are easily influenced by interfering items.

Attention and recurrent neural networks (RNNs) based SBR methods. Then, attention and RNNs are applied to SBRs. The attention-based SBR methods [14–16] can overcome this problem by using attention mechanisms to focus more on the items relevant to the target item and user preferences, but these methods ignore the sequential transition patterns between items in the session. Some RNNs-based SBR methods [17–19] use gated recurrent units (GRUs) to capture the sequential information of sessions. However, these methods encode the session as a simple unidirectional sequence and cannot capture more complex patterns of item transitions. Recently, for the sake of efficiency, Zhang et al. [20] removes the propagation part of GNN, and introduces a multi-level reasoning component in the attention-based readout module. However, this comes at the cost of sacrificing the strict relative order between items.

Graph neural networks (GNNs) based SBR methods. Recent research has widely adopted GNNs in SBRs, which can capture complex and high-order item transition relationships. SR-GNN [21] model

¹ <https://github.com/lean030/MLEUP>

was the first to use graph neural networks, integrating neighboring information of items using a gated graph neural network (GGNN). Qiu et al. [22] proposed a method that uses a weighted graph attention network to encode the neighboring information of items and studies the inherent order of item transition patterns. Wang et al. [23] captured multi-level user preferences by applying sliding windows of different sizes on the session sequence. Ouyang et al. [24] proposed to mine the changes of users' interest trends by learning multiple session embeddings with different lengths, and they put forward an adaptive loss function, which makes difficult samples get more attention. Li et al. [25] decouples item embeddings and explores user's finer-granularity interests at the factor level to generate session embeddings. However, the above methods are limited to capturing item transition information within a single session, without considering global information in historical sessions, leading to the serious problem of data sparsity. GCE-GNN [5] model was the first to use global level item transformation, which improved the recommendation performance by using pairwise item transfer information from two levels of session graph and global graph. Han et al. [26] proposed for the first time to learn multi-faceted global item relation, including a negative relationship that has not been studied before. These methods introduce global information but not fully take advantage of them and ignore attribute knowledge, still face the problem of data sparsity. Unlike this paper, Chen et al. [27] independently models attribute-related pattern and sequence-related pattern, enabling the attribute module to be plug-and-play. However, it overlooks the mutual influence between the two and restricts the mining of sequence patterns to a single session, thus failing to use sufficient information in the sequence-related pattern.

Self-supervised learning (SSL)/contrastive learning (CL) enhanced SBR methods. SSL can mine the feature representation of data itself as supervision signal, while CL can learn representation by constructing positive and negative samples. They can better solve the problem of data sparsity and are applied to SBRs recently. Meng et al. [28] proposed a SSL task, which uses TransH mechanism to incorporate item attribute knowledge into the model. Lai et al. [9] constructed an attribute-aware mirror graph based on session graph to aggregate items with the same attributes. They also designed an iterative refinement process to fully integrate items and item attributes, while utilizing a SSL task to supervise the generation process of the mirror graph. Cao et al. [29] proposed a neighbor contrast loss function to learn the inter-session item representation, which makes the computational efficiency and robustness of updating item features stronger. Xia et al. [30] proposed a model-independent joint framework integrating self-monitoring task, difference constraint task and recommendation task, which can be adapted to more scenarios. However, these methods only consider the first-order neighbor relationship and rely on stacking multiple layers of neural networks to incorporate high-order information, which can lead to over-smoothing [31]. Some studies have avoided this problem. For example, Xia et al. [3] proposed a dual-channel hypergraph convolution network, which can capture the beyond-pairwise relations among items through hypergraph modeling, and enhance hypergraph modeling by using SSL tasks. Tai et al. [32] maximized the mutual information between the current session embedding and the similar session embedding with the same target item through CL, so as to capture the intrinsic correlation between user behaviors. Peintner et al. [33] proposed a CL task, which optimizes the negative sample sampling method by using the modified BLEU score session similarity calculation strategy. But these methods only model one of the co-occurrence relationship and sequence relationship, and cannot distinguish the differences between the two relationships.

3. Problem definition

Given a chronologically ordered list of anonymous sessions from all users $X = [S_1, S_2, \dots, S_{|X|}]$, where $|X|$ denotes the number of sessions in X . For each session, it is represented as a chronological list of items

from an anonymous user $S_x = [v_1, v_2, \dots, v_{|S_x|}]$, which may contain duplicate items, where $|S_x|$ is the length of the session. Given an attribute set $C = \{c_1, c_2, \dots, c_{|C|}\}$ containing all attribute types, where $|C|$ represents the number of attribute categories. For each attribute c_n , its set of attribute values is represented as $A^{c_n} = \{a_1^n, a_2^n, \dots, a_{|A^{c_n}|}^n\}$, where $|A^{c_n}|$ denotes the number of attribute values for the corresponding attribute. Given an item set $I = \{v_1, v_2, \dots, v_{|I|}\}$ containing all the items, where $|I|$ denotes the number of items in I , each item v_i contains several attribute values $A^{v_i} = \{b_1^i, b_2^i, \dots, b_{|C|}^i\}$ and i denotes item ID. Given an ongoing session denoted as $S = [v_1, v_2, \dots, v_l]$, the goal of SBR is to recommend the item most likely to be clicked by the user at next time, i.e., v_{l+1} .

We list the main notations used in this article in Table 1.

4. The proposed MLEUP model

The proposed MLEUP is illustrated in Fig. 1. It consists of five components: (1) Session group construction. Given a session, the most similar sessions are selected from its historical session set to construct the session group. (2) Attribute embedding learning. This component learns the embeddings of attributes by modeling the global attribute-item relationship and the intra-session attribute-attribute relationship. (3) Item embedding learning. Taking the session group as input, this component learns the multi-granularity co-occurrence relationship of items through a weighted hypergraph attention (WHGAT) layer and the high-order sequential relationship among items through a relative position enhanced graph attention (RPGAT) layer. (4) User preferences extraction and fusion. This component first learns three types of session embeddings, and then generates dynamic and stable preferences of the user. By fusing these two types of preferences, it models the complex interaction between them and enhances their semantic meaning. (5) Multi-task joint training. We utilize the correlation between sessions and the attribute embeddings of sessions to design a neighbor-based contrastive self-supervised attribute learning task. Together with the recommendation task, they form a multi-task joint training framework. In the following sections, we will elaborate on each component in detail.

4.1. Session group construction

In this section, we introduce our proposed strategy for measuring session similarity. Firstly, we provide a definition of the cross-session item incompatibility relationship and highlight the relationship between cross-session item incompatibility and overlap with session similarity. Then, we outline the calculation process for determining the incompatibility relationship between items and measuring the similarity between sessions.

Inspired by [26], we can define an incompatibility relationship between two items that often co-occur with other items in sessions but do not co-occur with each other in the same session. For example, with "Apple computer" and "Lenovo computer", users usually do not purchase another computer after buying one. It is easy to conclude that the incompatibility relationship between items is a type of implicit negative relationship. The higher the percentage of item pairs that have an incompatibility relationship between two sessions, the lower the similarity between the sessions. Conversely, the overlap of items between sessions is a type of explicit positive relationship, and the higher the percentage of overlapping items between two sessions, the higher the similarity between the sessions. Therefore, we can calculate the similarity between sessions using these two relationships. The specific steps are as follows:

Firstly, find pairs of items that have an incompatibility relationship. As there are many long-tail items in the item set that have little interaction, they can lead to many erroneous incompatibility relationships. We count the frequency of items appearing in the session set and define a threshold hyperparameter $\delta \in [0, 1]$. We take the lowest δ proportion of

Table 1
Main notations used in this article.

Notation	Description
X	the anonymous session set containing $ X $ chronological sessions from all users
S_x	an anonymous session containing $ S_x $ chronological items
C, I	the attribute set and item set
c_n, v_i	the n th attribute, the i th item
A^c, A^v	the attribute values sets of attribute c_n and item v_i
a_o^n, b_c^j	the o th attribute value of attribute c_n , the c th attribute value of item v_i
S, G_S	the ongoing session and its session group
m	the number of sessions in G_S
$N(a_o^n)$	the sample set of items whose attribute value is a_o^n
h_{v_i}	the initial embedding of item v_i
$h_{a_o^n}, h_{b_c^j}$	the initial embedding of attribute value a_o^n , the information of the items associated with a_o^n
$g_{a_o^n}, g_{b_c^j}, g_{v_i}$	the global attribute-item relationship embeddings of attribute value a_o^n , b_c^j and item v_i
P^+, P^-	the forward and backward position embedding matrixes
$G_S^* = [g_{v_1}^*, g_{v_2}^*, \dots, g_{v_i}^*]^T$	the position encoded global attribute-item relationship attribute embedding matrix of session S
$G_S^{att} = [g_{v_1}^{att}, g_{v_2}^{att}, \dots, g_{v_i}^{att}]^T$	the final attribute embedding matrix of session S
$G_{S \cup G_S}^{co}, G_{S \cup G_S}^{seq}$	the weighted session hypergraph and heterogenous graph
h_e, w_{e_j}	the feature vector and the weight corresponding to hyperedge e_j
$H_S^{co} = [h_{v_1}^{co}, h_{v_2}^{co}, \dots, h_{v_i}^{co}]^T$	the co-occurrence relationship embedding matrix of session S
$N_{1 \sim \theta}^+(v_i), N_{1 \sim \theta}^-(v_i)$	the $1 \sim \theta$ -th order both forward and backward neighboring items in $S \cup G_S$ of item v_i
$N_k^+(v_i), N_k^-(v_i)$	the k th order forward and backward neighboring items in $S \cup G_S$ of item v_i
$h_{v_i}^+, h_{v_i}^-$	the forward and backward position encoded embeddings of item v_i
$m_{v_i}^+, m_{v_i}^-, m_{v_i}$	the forward, backward and final high-order neighbor information of item v_i
$w_{e_{i,j}}, w_{e_{j,i}}^{seq}, w_{e_{i,j}}^{seq}, w_{e_{j,i}}^{seq}$	the weights of edge $e_{i,j}$ and $e_{j,i}$
$H_S^{seq} = [h_{v_1}^{seq}, h_{v_2}^{seq}, \dots, h_{v_i}^{seq}]^T$	the sequential relationship embedding matrix of session S
s^{att}, s^{co}, s^{seq}	the attribute, co-occurrence and sequential embeddings of session S
s^s, s^d	the user's stable and dynamic preferences
$s^{s'}, s^{d'}$	the fused user's stable and dynamic preferences

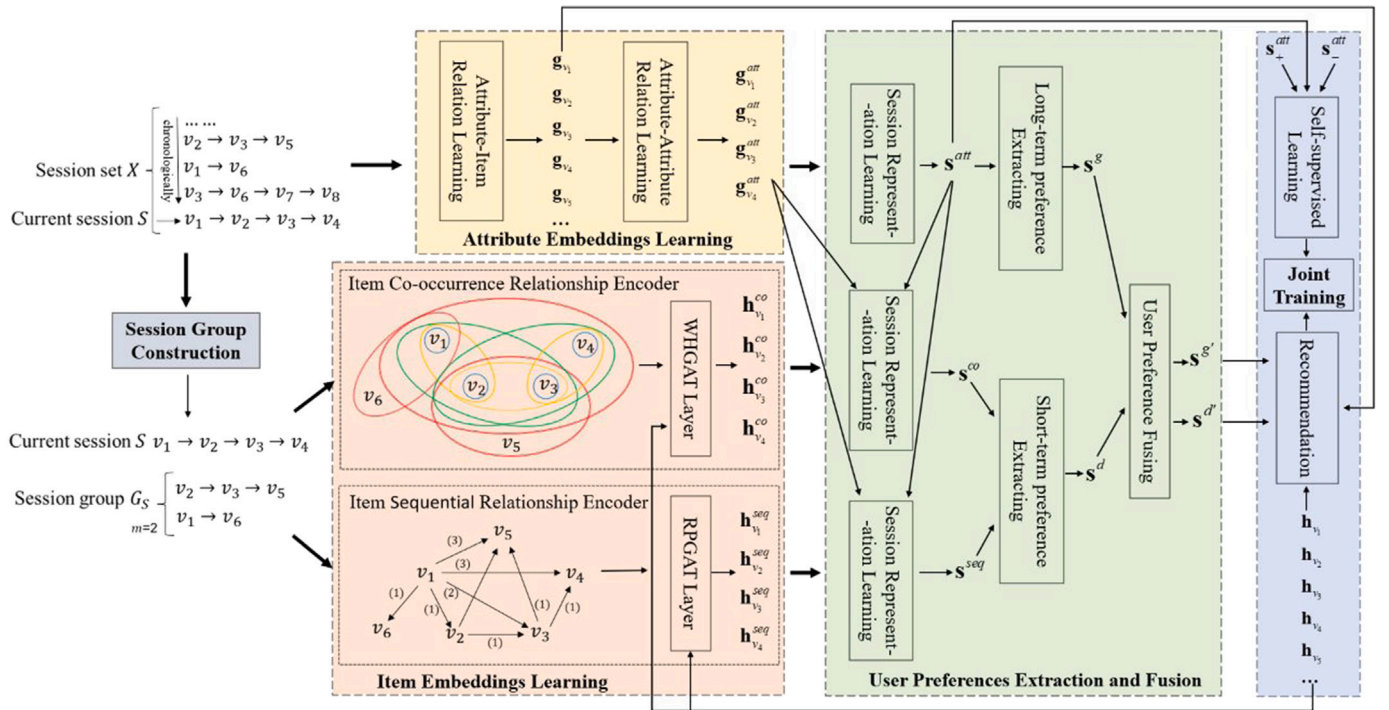


Fig. 1. Overview of the MLEUP framework.

items based on frequency as the long-tail item set $\bar{I}$ and exclude them from counting incompatibility relationships. Then we calculate the co-occurrence frequency of two items v_i and v_j in the set of sessions, which is defined as $p_{i,j} = \sum_{x=1}^{|X|} 1_{v_i \in S_x \& v_j \in S_x}$, where $1_{v_i \in S_x \& v_j \in S_x}$ is an indicator function that equals to 1 when v_i and v_j co-occur in a session and 0 otherwise, and $|X|$ represents the number of sessions in the set X . We then define a threshold parameter $\mu > 0$. If $p_{i,j} = 0$ and there exists an item v_k such that $p_{i,k}$ and $p_{j,k}$ both reach the threshold μ , we consider

that there is an incompatibility relationship between v_i and v_j , denotes as $v_i \nabla v_j$.

Secondly, calculate the similarity between two sessions. It can be obtained by computing the difference between their overlap and incompatibility as follows:

$$Sim(S_x, S_y) = Ov(S_x, S_y) - In(S_x, S_y) \quad (1)$$

Considering the possibility of duplicate items within a session and the differences in overlap and incompatibility relationships, we design different algorithms to calculate them. Specifically, the overlap between two sessions is calculated as the ratio of the number of overlapping item pairs to the length of the longer session, as follows:

$$Ov(S_x, S_y) = \frac{\sum_{i=1}^{\min(|S_x|, |S_y|)} 1_{v_i^{\text{sorted}(S_x)} = v_i^{\text{sorted}(S_y)}}}{\max(|S_x|, |S_y|)} \quad (2)$$

where $\text{sorted}(S_x)$ and $\text{sorted}(S_y)$ denotes that the items in the session S_x and S_y are arranged in ascending order by the item ID, which is the number of the item in I , $1_{v_i^{\text{sorted}(S_x)} = v_i^{\text{sorted}(S_y)}}$ is an indicator function that equals to 1 if the same item appears in the same position in the sorted sequences of the two sessions, and 0 otherwise.

The incompatibility between two sessions is calculated as the ratio of the number of incompatible item pairs between the sessions to the product of their lengths, as follows:

$$In(S_x, S_y) = \frac{\sum_{i=1}^{|S_x|} \sum_{j=1}^{|S_y|} 1_{v_i \nabla v_j}}{|S_x| \times |S_y|} \quad (3)$$

where $1_{v_i \nabla v_j}$ is an indicator function that equals to 1 if the item at a certain position in one session is incompatible with the item at a certain position in the other session, and 0 otherwise.

Finally, we define a threshold hyperparameter $m > 0$ and choose the m historical sessions with the highest similarity, which forms the session group G_S of a given session S .

4.2. Attribute embedding learning

In this section, we introduce the process of learning attribute embeddings by modeling attribute-item and attribute-attribute relationships. Usually, an item in a dataset contains multiple attributes, and each attribute value is associated with multiple items, thus forming a correlation between attributes and items. In addition, the attributes of items in a session are also interrelated. Therefore, we learn the attribute embeddings of the items by modeling these two relationships.

Attribute-item relationship. We set a hyperparameter ω , and for each attribute value $a_o^n \in A^{c_n}$ of attribute $c_n \in C$, we sample a ω proportion of items that have the corresponding attribute value from the session set, forming a set of items $N(a_o^n)$. We first distinguish the importance of different items using the attention mechanism as follows:

$$\varphi_i = \text{softmax}(\mathbf{q}^{\text{att}} \mathbf{h}_{v_i}) \quad (4)$$

$$\mathbf{h}'_{a_o^n} = \sum_{v_i \in N(a_o^n)} \varphi_i \mathbf{h}_{v_i} \quad (5)$$

where $\mathbf{q}^{\text{att}} \in \mathbb{R}^d$ is a trainable parameter, d is the embedding size, $\mathbf{h}_{v_i}$ denotes the initial embedding of item v_i , and $\mathbf{h}'_{a_o^n}$ denotes the information of the items associated with attribute value a_o^n . Then, in order to alleviate the over-fitting problem caused by multi-layer network training, we apply the highway network [34] to aggregate the initial embedding $\mathbf{h}_{a_o^n}$ of attribute value a_o^n and the corresponding item information $\mathbf{h}'_{a_o^n}$. It uses fewer parameters and can be chosen to learn useful information from different layers, as follows:

$$\mathbf{g}_{a_o^n} = \text{highway}(\mathbf{h}_{a_o^n}, \mathbf{h}'_{a_o^n}) \quad (6)$$

where $\mathbf{g}_{a_o^n}$ denotes the updated embedding vector of attribute value a_o^n , and the definition of the $\text{highway}(\mathbf{h}_1, \mathbf{h}_2)$ is as follows:

$$\pi = \sigma(\mathbf{W}[\mathbf{h}_1 \parallel \mathbf{h}_2]) \quad (7)$$

$$\mathbf{h}_{\text{final}} = \pi \odot \mathbf{h}_1 + (1 - \pi) \odot \mathbf{h}_2 \quad (8)$$

where $\mathbf{W} \in \mathbb{R}^{d \times 2d}$ is a trainable parameter, $|\cdot|$ represents the concatenation operation, $\odot$ denotes the Hadamard product, $\sigma(\cdot)$ denotes the

sigmoid function, $\mathbf{h}_1$ and $\mathbf{h}_2 \in \mathbb{R}^d$ denotes the embedding vector, and $\mathbf{h}_{\text{final}} \in \mathbb{R}^d$ denotes the updated embedding vector.

Therefore, we obtain the embedding matrix for each attribute c_n . Finally, we select the embedding vector of the attribute value corresponding to item v_i from each attribute's embedding matrix and concatenate them together as follows:

$$\mathbf{g}_{v_i} = \mathbf{W}_1^{\text{att}} [\mathbf{g}_{b_1^i} \parallel \mathbf{g}_{b_2^i} \parallel \cdots \parallel \mathbf{g}_{b_{|C|}^i}] \quad (9)$$

where $\mathbf{W}_1^{\text{att}} \in \mathbb{R}^{d \times |C|d}$ is a trainable parameter, and $\mathbf{g}_{v_i}$ denotes the attribute embedding of item v_i , which only contains the global attribute-item relationship. It is easy to see that the number of attributes can be freely expanded.

Attribute-attribute relationship. To learn the attribute-attribute relationship within a session, we first concatenate a position embedding with the attribute embedding for each item in the session as follows:

$$\mathbf{g}_{v_i}^* = \tanh(\mathbf{W}_2^{\text{att}} [\mathbf{g}_{v_i} \parallel \mathbf{p}_{l-i+1}^+]) \quad (10)$$

where $\mathbf{W}_2^{\text{att}} \in \mathbb{R}^{d \times 2d}$ is a trainable parameter, l is the length of the session, and $\mathbf{p}_{l-i+1}^+$ is the forward position embedding, which is initialized by the lookup of the forward position embedding matrix $\mathbf{P}^+$, because other items within the session are forward-looking items with respect to the target item. Therefore, we obtain the attribute embedding matrix $\mathbf{G}_S^* = [\mathbf{g}_{v_1}^*, \mathbf{g}_{v_2}^*, \dots, \mathbf{g}_{v_l}^*]^T$ with concatenated positional embeddings for session S . Then, we apply a multi-head self-attention mechanism that propagates information among the attribute embeddings of the items within the session as follows:

$$\text{head}_i = \text{softmax}\left(\frac{(\mathbf{G}_S^* \mathbf{W}_{Q_i})(\mathbf{G}_S^* \mathbf{W}_{K_i})^T}{\sqrt{\frac{d}{h}}}\right)(\mathbf{G}_S^* \mathbf{W}_{V_i}) \quad (11)$$

$$\mathbf{G}_S^{\text{att}} = [\text{head}_1 \parallel \text{head}_2 \parallel \cdots \parallel \text{head}_h] \quad (12)$$

where $\mathbf{W}_{Q_i}$, $\mathbf{W}_{K_i}$ and $\mathbf{W}_{V_i} \in \mathbb{R}^{d \times \frac{d}{h}}$ are trainable parameters, h denotes the number of attention heads, and $\mathbf{G}_S^{\text{att}} = [\mathbf{g}_{v_1}^{\text{att}}, \mathbf{g}_{v_2}^{\text{att}}, \dots, \mathbf{g}_{v_l}^{\text{att}}]^T$ denotes the final sequence of attribute embeddings for the original session S .

4.3. Item embedding learning

4.3.1. Item co-occurrence relationship encoder

In this section, we introduce the encoding process of the item co-occurrence relationship in the session. First, we construct a weighted session hypergraph based on the session itself and its session group. Then, we design a WHGAT layer to learn the multi-granularity co-occurrence relationship information propagated in the session hypergraph graph.

Session hypergraph construction. Given session S and its session group $G_S = \{S_1, S_2, \dots, S_m\}$, we can use sliding windows of different sizes to collect item co-occurrence relationship with multiple levels of granularity, and the construction process of the weighted session hypergraph is as shown in Fig. 2.

Specifically, we use $\mathcal{G}_{S \cup G_S}^{\text{co}} = (\mathcal{V}_{S \cup G_S}^{\text{co}}, \mathcal{E}_{S \cup G_S}^{\text{co}})$ to represent the weighted session hypergraph, where the node set $\mathcal{V}_{S \cup G_S}^{\text{co}}$ includes all items that appear in the session S and its session group G_S , and the hyperedge set $\mathcal{E}_{S \cup G_S}^{\text{co}}$ is divided into two parts: (1) The multi-granularity hyperedge set $\mathcal{E}_S^{\text{co}}$ generated based on session S . By changing the size of the sliding window and aggregating the hyperedges of different sizes, we obtain the hyperedge set $\mathcal{E}_S^{\text{co}} = \mathcal{E}_{S,l}^{\text{co}} \cup \mathcal{E}_{S,l-1}^{\text{co}} \cup \mathcal{E}_{S,l-2}^{\text{co}} \cup \cdots \cup \mathcal{E}_{S,z}^{\text{co}}$, where $z < l$ is the maximum length of the sliding window chosen. (2) The hyperedge set $\mathcal{E}_{G_S}^{\text{co}}$ generated based on the session group G_S . For each session in the session group, we generate a hyperedge by taking the complete sequence of the session, and then aggregate them to obtain the hyperedge set $\mathcal{E}_{G_S}^{\text{co}} = \mathcal{E}_{S_1,l_1}^{\text{co}} \cup \mathcal{E}_{S_2,l_2}^{\text{co}} \cup \cdots \cup \mathcal{E}_{S_m,l_m}^{\text{co}}$. Note that we remove duplicate items in the hyperedges and delete duplicate hyperedges. Intuitively, it is believed that different time spans of other sessions from the current session S may provide different levels of importance to the

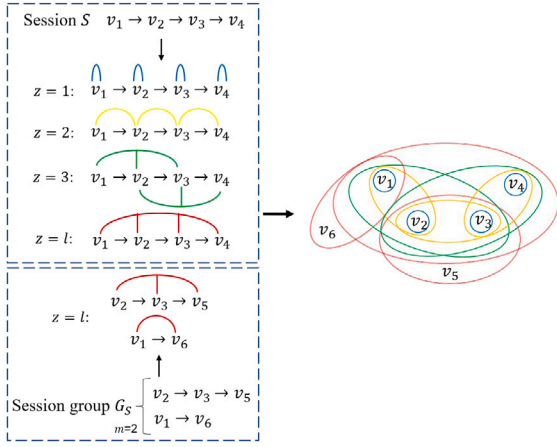


Fig. 2. Example of the weighted session hypergraph construction process.

information. Therefore, we generate weights for each hyperedge e as $w_e = \frac{1}{2^{t-t'}}$, where t is the timestamp of session S and t' is the timestamp of the session where hyperedge e comes from.

WHGAT layer. Inspired by [35], the learning process we designed consists of node-to-hyperedge and hyperedge-to-node propagation.

First, node-to-hyperedge propagation. In the constructed session hypergraph, hyperedges simulate user intentions at different granularities, but some nodes in the hyperedge may introduce irrelevant noise information. Therefore, we use attention mechanisms to aggregate the node embeddings $\mathbf{h}_{v_i}$ contained in hyperedge e_j . Each hyperedge e_j is treated as a cluster of the item nodes linked by the hyperedge, and the center of the cluster is calculated as $\mathbf{h}_{e_j}^* = \text{mean}(\mathbf{h}_{v_i} | v_i \in e_j)$. Since nodes closer to the center of the cluster are more likely to be primary intentions, the attention process is designed as follows:

$$\alpha_{ji} = \text{softmax}(f(\mathbf{q}^{coT}(\mathbf{h}_{e_j}^* \odot \mathbf{h}_{v_i}))) \quad (13)$$

$$\mathbf{h}_{e_j} = \sum_{v_i \in e_j} \alpha_{ji} \mathbf{h}_{v_i} \quad (14)$$

where $\mathbf{q}^{co} \in \mathbb{R}^d$ is a trainable parameter, $f(\cdot)$ denotes the *LeakyReLU* function, and $\mathbf{h}_{e_j}$ represents the feature vector corresponding to hyperedge e_j .

Second, hyperedge-to-node propagation. After obtaining the features of the hyperedges, we apply a weighted attention mechanism to incorporate the pre-defined hyperedge weights as follows:

$$\mathbf{h}_{v_i}^{co} = \text{weightedatt}(\mathbf{h}_{v_i}, \mathbf{h}_{e_j}, w_{e_j}) \quad (15)$$

where e_j is from the set of hyperedges that contains the item node v_i , w_{e_j} is the weight of the hyperedge e_j , $\mathbf{h}_{v_i}^{co}$ denotes the co-occurrence relationship embedding of item v_i in the session, and the definition of the *weightedatt*($\mathbf{g}, \mathbf{g}_j, w_j$) is as follows:

$$\beta_j = \text{softmax}(\mathbf{q}^T f(\mathbf{W}[(\mathbf{g} \odot \mathbf{g}_j) \parallel w_j])) \quad (16)$$

$$\mathbf{g}_{final} = \sum_j \beta_j \mathbf{g}_j \quad (17)$$

where $\mathbf{W} \in \mathbb{R}^{d \times (d+1)}$ and $\mathbf{q} \in \mathbb{R}^d$ are trainable parameters, $\mathbf{g} \in \mathbb{R}^d$ is the query vector, $\mathbf{g}_j \in \mathbb{R}^d$ is the key and value vector, w_j is the weight value, and $\mathbf{g}_{final} \in \mathbb{R}^d$ is the updated embedding vector.

With the co-occurrence relationship encoder, we can learn a item co-occurrence relationship embedding matrix for the session without duplicate items, and thus obtain a sequence of item co-occurrence relationship embeddings $\mathbf{H}_S^{co} = [\mathbf{h}_{v_1}^{co}, \mathbf{h}_{v_2}^{co}, \dots, \mathbf{h}_{v_l}^{co}]^T$ for the original session S .

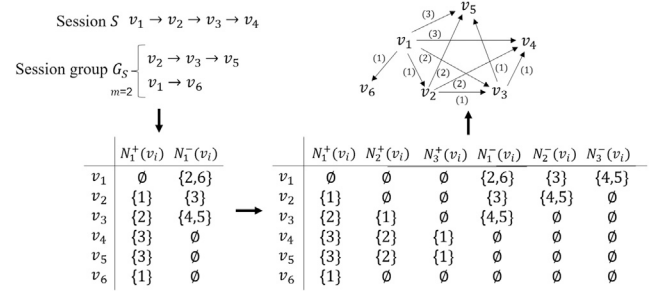


Fig. 3. Example of the weighted session heterogeneous graph construction process.

4.3.2. Item sequential relationship encoder

In this section, we introduce the encoding process of the item sequential relationship in the session. First, we construct a weighted session heterogeneous graph based on the session itself and its session group. Then, we design a RPGAT layer to learn the high-order sequential relationship information propagated in the session heterogeneous graph.

Session heterogeneous graph construction. Firstly, we define the set of high-order neighboring items for a item. Give a session S and its session group G_S , we collect the high-order neighboring items $N_{1 \sim \theta}(v_i)$ for each item v_i in session S within the range of θ before and after the session set $S \cup G_S$. Note that when the item v_j appears in the multi-order neighbor range of v_i , we take the minimum order.

In the session, when there is a pairwise item transition from v_i to v_j , v_i is the first-order forward neighbor of v_j , and v_j is the first-order backward neighbor of v_i . Therefore, we first collect the first-order forward neighbor set $N_1^+(v_i)$ and first-order backward neighbor set $N_1^-(v_i)$ for each item v_i in the session set $S \cup G_S$. Then, for each item v_i in the session S , its k -order forward neighbor set and k -order backward neighbor set are recursively defined as follows:

$$N_k^+(v_i) = \bigcup_{v_j \in N_{k-1}^+(v_i)} \{v_k | v_k \in N_1^+(v_j); v_k \notin N_{1 \sim k-1}^+(v_i)\} \quad (18)$$

$$N_k^-(v_i) = \bigcup_{v_j \in N_{k-1}^-(v_i)} \{v_k | v_k \in N_1^-(v_j); v_k \notin N_{1 \sim k-1}^-(v_i)\} \quad (19)$$

Secondly, we construct the weighted session heterogeneous graph, as shown in Fig. 3. Specifically, we use $\mathcal{G}_{S \cup G_S}^{\text{seq}} = (\mathcal{V}_{S \cup G_S}^{\text{seq}}, \mathcal{E}_{S \cup G_S}^{\text{seq}})$ to represent the weighted session heterogeneous graph, where the node set $\mathcal{V}_{S \cup G_S}^{\text{seq}}$ includes each item in session S and its neighbors in $N_{1 \sim \theta}(v_i)$ collected within the session set $S \cup G_S$, and the edge set $\mathcal{E}_{S \cup G_S}^{\text{seq}}$ is divided into θ types based on the different distances and positional relationship between the pairs of neighboring items: when v_j is a k -order neighbor of v_i , if v_i points to v_j , there is an edge $e_{i,j}^k$ connecting them, and if v_j points to v_i , there is an edge $e_{j,i}^k$ connecting them. At the same time, we generate a weight for each edge. For edge $e_{i,j}^k$, its weight is defined as $w_{e_{i,j}^k} = \sum_{S' \in S \cup G_S} \frac{p_{i,j}^k}{2^{t-t'}}$, where t represents the timestamp of session S , t' represents the timestamp of the session S' that v_j comes from, and $p_{i,j}^k$ represents the frequency with which v_j appears as a k -order backward neighbor of v_i in session S' . Similarly, for edge $e_{j,i}^k$, its weight is defined as $w_{e_{j,i}^k} = \sum_{S' \in S \cup G_S} \frac{p_{j,i}^k}{2^{t-t'}}$, where $p_{j,i}^k$ represents the frequency with which v_j appears as a k -order forward neighbor of v_i in session S' .

RPGAT layer. The specific steps we designed are as follows:

Firstly, centering around the item node v_i , incorporate the positional information of its neighboring nodes using the shared relative positional embedding matrix. For the forward neighboring node $v_j \in N_{1 \sim \theta}(v_i)$ of node v_i , the forward position embedding $\mathbf{p}_{1 \sim \theta}^+$ is concatenated with the neighbor node embedding $\mathbf{h}_{v_j}$, and the same applies to the backward neighbors of node v_i , as follows:

$$\mathbf{h}_{v_j}^+ = \tanh(\mathbf{W}^+[\mathbf{h}_{v_j} \parallel \mathbf{p}_{1 \sim \theta}^+]) \quad (20)$$

$$\mathbf{h}_{v_j}^- = \tanh(\mathbf{W}^+[\mathbf{h}_{v_j} \parallel \mathbf{p}_{i-j}^-]) \quad (21)$$

where $\mathbf{W}^+$ and $\mathbf{W}^- \in \mathbb{R}^{d \times 2d}$ are trainable parameters. Same as $\mathbf{p}^+$, $\mathbf{p}^-$ is initialized by the lookup of the backward position embedding matrix $\mathbf{P}^-$.

Secondly, learn the information propagated from high-order adjacent edges of node v_i . We first apply the weighted attention mechanism to calculate the information propagated from the forward and backward neighbors separately, as follows:

$$\mathbf{m}_{v_i}^+ = \text{weightedatt}(\mathbf{h}_{v_i}, \mathbf{h}_{v_j}^+, w_{e_{j,i}}) \quad (22)$$

$$\mathbf{m}_{v_i}^- = \text{weightedatt}(\mathbf{h}_{v_i}, \mathbf{h}_{v_j}^-, w_{e_{i,j}}) \quad (23)$$

where $w_{e_{j,i}}$ is the weight of the edge $e_{j,i}$, $w_{e_{i,j}}$ is the weight of the edge $e_{i,j}$, $\mathbf{m}_{v_i}^+$ and $\mathbf{m}_{v_i}^-$ denote the forward and backward high-order neighbor information, respectively. Then, by aggregating them, the final neighbor information $\mathbf{m}_{v_i}$ of node v_i is obtained as follows:

$$\mathbf{m}_{v_i} = \mathbf{m}_{v_i}^+ + \mathbf{m}_{v_i}^- \quad (24)$$

where $+$ denotes vector element-wise addition.

Thirdly, update node embedding of v_i . We apply the highway network to selectively obtain information from the node's initial embedding $\mathbf{h}_{v_i}$ and its neighbor feature embedding $\mathbf{m}_{v_i}$ as follows:

$$\mathbf{h}_{v_i}^{seq} = \text{highway}(\mathbf{h}_{v_i}, \mathbf{m}_{v_i}) \quad (25)$$

where $\mathbf{h}_{v_i}^{seq}$ denotes the sequential relationship embedding of item v_i in the session.

With the sequential relationship encoder, we can learn a sequential relationship embedding matrix for the session without duplicate items, and thus obtain a sequence of sequential relationship embeddings $\mathbf{H}_S^{seq} = [\mathbf{h}_{v_1}^{seq}, \mathbf{h}_{v_2}^{seq}, \dots, \mathbf{h}_{v_l}^{seq}]^T$ for the original session S .

4.4. User preferences extraction and fusion

In this section, we first extract stable and dynamic user preferences by learning the attribute embedding of the session and two types of relationship embeddings of the session. We then adopt a co-guided learning mechanism to model the interaction between the two, allowing them to integrate with each other and guide each other's learning process.

4.4.1. User preferences extraction

Stable user preference extraction. As previously mentioned, we learn the attribute embedding representation of session, denoted as $\mathbf{s}^{att}$, as the user's stable preference. Specifically, we use a simple attention network to aggregate the attribute feature of each item, as follows:

$$\gamma_i = \text{softmax}(\mathbf{q}_1^T \mathbf{g}_{v_i}^{att}) \quad (26)$$

$$\mathbf{s}^{att} = \sum_{i=1}^l \gamma_i \mathbf{g}_{v_i}^{att} \quad (27)$$

where $\mathbf{q}_1 \in \mathbb{R}^d$ is a trainable parameter, and $\mathbf{s}^g = \mathbf{s}^{att}$ denotes the user's stable preference.

Dynamic user preference extraction. The two types of relationship embeddings learned from the session can reflect the user's dynamic preference, while we introduce the position embeddings to capture the user's dynamic interests that change over time. Since other items within the session are forward-looking items with respect to the target item, we reuse the forward position embedding matrix. Specifically, we first combine the position embedding with the two types of item relationship embedding, as follows:

$$\mathbf{h}_{v_i}^{co*} = \tanh(\mathbf{W}_1[\mathbf{h}_{v_i}^{co} \parallel \mathbf{p}_{l-i+1}^+]) \quad (28)$$

$$\mathbf{h}_{v_i}^{seq*} = \tanh(\mathbf{W}_2[\mathbf{h}_{v_i}^{seq} \parallel \mathbf{p}_{l-i+1}^+]) \quad (29)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2 \in \mathbb{R}^{d \times 2d}$ are trainable parameters.

After that, we introduce the item attribute embedding $\mathbf{g}_{v_i}^{att}$ to enrich the item information and introduce the session attribute embedding $\mathbf{s}^{att}$ as the user's general interest to guide the generation of the session embedding about item relationships, then apply the attention mechanism as follows:

$$\theta_i^{co} = \text{softmax}(\mathbf{q}_2^T \sigma(\mathbf{W}_3 \mathbf{h}_{v_i}^{co*} + \mathbf{W}_4 \mathbf{g}_{v_i}^{att} + \mathbf{W}_5 \mathbf{s}^{att} + b_1)) \quad (30)$$

$$\mathbf{s}^{co} = \sum_{i=1}^l \theta_i^{co} \mathbf{h}_{v_i}^{co*} \quad (31)$$

$$\theta_i^{seq} = \text{softmax}(\mathbf{q}_3^T \sigma(\mathbf{W}_6 \mathbf{h}_{v_i}^{seq*} + \mathbf{W}_7 \mathbf{g}_{v_i}^{att} + \mathbf{W}_8 \mathbf{s}^{att} + b_2)) \quad (32)$$

$$\mathbf{s}^{seq} = \sum_{i=1}^l \theta_i^{seq} \mathbf{h}_{v_i}^{seq*} \quad (33)$$

where $\mathbf{W}_3, \mathbf{W}_4, \mathbf{W}_5, \mathbf{W}_6, \mathbf{W}_7$ and $\mathbf{W}_8 \in \mathbb{R}^d$, $\mathbf{q}_2$ and $\mathbf{q}_3 \in \mathbb{R}^d$ are trainable parameters.

Finally, we adopt the gating mechanism to integrate the two and learn the comprehensive session embedding about item relationships as the user's dynamic preference, as follows:

$$\mathbf{s}^d = \text{highway}(\mathbf{s}^{co}, \mathbf{s}^{seq}) \quad (34)$$

where $\mathbf{s}^d$ denotes the user's dynamic preference.

4.4.2. User preferences fusion

The user's stable preference generated by the attribute information of items and the user's dynamic preference generated by the relationship information among items interact with each other and jointly determine user choices. Therefore, inspired by [8], we adopt a co-guided learning mechanism to model the interaction between the two.

We first integrate the stable preference with the dynamic preference in two ways:

$$\mathbf{i}_1 = \tanh(\mathbf{W}_9(\mathbf{s}^g \odot \mathbf{s}^d) + b_3) \quad (35)$$

$$\mathbf{i}_2 = \tanh(\mathbf{W}_{10}(\mathbf{s}^g + \mathbf{s}^d) + b_4) \quad (36)$$

where $\mathbf{W}_9$ and $\mathbf{W}_{10} \in \mathbb{R}^{d \times d}$ are trainable parameters, $\mathbf{i}_1$ and $\mathbf{i}_2 \in \mathbb{R}^d$ represent interactive relations between stable and dynamic preferences in different semantic spaces.

Then, we apply the gating mechanism to further model the mutual relations between $\mathbf{s}^g$ and $\mathbf{s}^d$ as follows:

$$\mathbf{z}_1 = \sigma(\mathbf{W}_{11}[\mathbf{i}_1 \parallel \mathbf{i}_2]) \quad (37)$$

$$\mathbf{z}_2 = \sigma(\mathbf{W}_{12}[\mathbf{i}_1 \parallel \mathbf{i}_2]) \quad (38)$$

$$\mathbf{r}^g = \tanh(\mathbf{W}_{13}(\mathbf{z}_1 \odot \mathbf{s}^g) + \mathbf{W}_{14}((1 - \mathbf{z}_2) \odot \mathbf{s}^d)) \quad (39)$$

$$\mathbf{r}^d = \tanh(\mathbf{W}_{15}((1 - \mathbf{z}_1) \odot \mathbf{s}^g) + \mathbf{W}_{16}(\mathbf{z}_2 \odot \mathbf{s}^d)) \quad (40)$$

where $\mathbf{W}_{11}$ and $\mathbf{W}_{12} \in \mathbb{R}^{d \times 2d}$, $\mathbf{W}_{13}$, $\mathbf{W}_{14}$, $\mathbf{W}_{15}$ and $\mathbf{W}_{16} \in \mathbb{R}^{d \times d}$ are trainable parameters.

Finally, we obtain the final user's stable and dynamic preferences as follows:

$$\mathbf{s}^{g'} = \mathbf{r}^g \odot (\mathbf{s}^g + \mathbf{r}^d) \quad (41)$$

$$\mathbf{s}^{d'} = \mathbf{r}^d \odot (\mathbf{s}^d + \mathbf{r}^g) \quad (42)$$

4.5. Multi-task joint training

In this section, we introduce the SSL task based on constructed session group and session attribute embeddings, which, together with the main recommendation task, forms a multi-task joint training framework.

4.5.1. Neighbor-based contrastive self-supervised attribute learning task

To fully leverage the information of the generated session group and improve the quality of attribute embeddings, thus assisting in learning more accurate user preferences, we designed a neighbor-based contrastive self-supervised attribute learning process to derive supervision signals from the correlations between sessions. Specifically, given a session S , it is more similar to the sessions in its session group G_S . The attribute embedding of session S and the attribute embedding of session from its session group G_S form a positive pair. Thus, we can generate m pairs of positive samples. At the same time, we collect m sessions from outside the session group, their attribute embeddings and the attribute embedding of session S form m pairs of negative samples. Then we adopt the contrastive loss function InfoNCE [36] to maximize the similarity between positive samples and minimize the similarity between negative samples, as follows:

$$\mathcal{L}_s = - \sum_{j=1}^m \log \sigma(f(s^{att}, s_{+j}^{att}) + \log \sigma(1 - f(s^{att}, s_{-j}^{att}))) \quad (43)$$

where $\langle s^{att}, s_{+j}^{att} \rangle$ forms a positive sample, $\langle s^{att}, s_{-j}^{att} \rangle$ forms a negative sample, and $f(\cdot, \cdot) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a discriminative function defined as $f(\mathbf{a}, \mathbf{b}) = \mathbf{a}^T \mathbf{b}$.

4.5.2. Main recommendation task

Give a session S , we calculate the score $\hat{y}_i$ for all candidate items based on the two user preferences ($s^{g'}$ and $s^{d'}$) and the two types of embeddings ($\mathbf{g}_{v_i}$ and $\mathbf{h}_{v_i}$) as follows:

$$\hat{y}_i = s^{g'T} \mathbf{g}_{v_i} + s^{d'T} \mathbf{h}_{v_i} \quad (44)$$

where $\mathbf{g}_{v_i}$ denotes the attribute embedding, and $\mathbf{h}_{v_i}$ denotes the initial item embedding. We then apply the *softmax* function to obtain the final scores as follows:

$$\hat{y}_i = \text{softmax}(\hat{y}_i) \quad (45)$$

After that, we adopt cross-entropy as the loss function:

$$\mathcal{L}_m = - \sum_{i=1}^{|I|} y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \quad (46)$$

where $|I|$ denotes the number of items in I , and y_i is the one-hot encoding vector of the ground truth.

After obtaining the recommendation task loss and the self-supervised task loss, we combine them as the final loss for joint training, as follows:

$$\mathcal{L} = \mathcal{L}_m + \lambda \mathcal{L}_s \quad (47)$$

where λ is used to control the strength of the self-supervised task. Finally, we apply the backpropagation through time (BPTT) algorithm to optimize the final loss $\mathcal{L}$.

5. Experiments

5.1. Datasets

We apply three real-world datasets to evaluate our model, known as Tmall,² Diginetica³ and 30music.⁴

Tmall dataset is from the IJCAI-15 competition, which contains users' shopping logs on Tmall in the past six months before and on the "Double 11" day. Each item includes three attributes (i.e., category, brand and seller).

Diginetica dataset comes from the Personalized E-commerce Search Challenge of CIKM Cup 2016, containing product transaction data from

Table 2

Statistics of the datasets.

Dataset	Tmall	Diginetica	30music
training sessions	351 268	719 470	223 222
test sessions	25 898	60 858	22 527
# items	40 727	43 097	36 979
# attribute num	3	2	1
avg. len.	6.69	4.85	6.31

an online retailer, and each item includes two attributes (i.e., category and price).

30music dataset collects listening and playlist data retrieved from Internet radio stations using the Last.fm API, where each item includes one attribute (i.e., artist).

Following [21,24,26], we preprocess three datasets as follows: Items appearing less than five times and sessions with only one item are filtered out. Due to the large scale of the Tmall and 30music datasets, we only take the first 120,000 records of them. Then, the most recent 100 s of sessions from the Tmall dataset, and the most recent 1 week of sessions from the Diginetica and 30music datasets are taken as the test set, while the remaining sessions are used for training. In addition, for every session $S = [v_1, v_2, \dots, v_n]$, we generate sequences and corresponding labels by a sequence splitting preprocessing, i.e., $([v_1], v_2), ([v_1, v_2], v_3), \dots, ([v_1, v_2, \dots, v_{n-1}], v_n)$ for both training and testing across all the three datasets. Table 2 summarizes the statistical data of the three datasets after preprocessing.

5.2. Evaluation metrics

Following the previous research work [5,8,21], we adopt three widely used ranking-based metrics.

P@K (Precision): It measures whether the ground-truth item is included in the top-K list.

$$P@K = \frac{n_{hit}}{|X_{test}|} \quad (48)$$

where X_{test} is the set of test sequences, and n_{hit} is the number of cases that the ground-truth item is in the top-K list.

M@K (Mean Reciprocal Rank): It indicates the position of the ground-truth item in the list of recommendation items. It is set to 0 if the ground-truth item is not in the top-K list, and otherwise is obtained as follows:

$$M@K = \frac{1}{|X_{test}|} \sum_{v_{gt} \in X_{test}} \frac{1}{Rank(v_{gt})} \quad (49)$$

where v_{gt} is the ground-truth item, and $Rank(v_{gt})$ is the position of the ground-truth item in the top-K list.

N@K (Normalized Discounted Cumulative Gain): It is another ranking-dependent metric of ranking quality in information retrieval tasks. It can be abbreviated in SBR as follows:

$$N@K = \frac{1}{|X_{test}|} \sum_{S \in X_{test}} \frac{2^{rel(v_{gt})} - 1}{\log_2(1 + Rank(v_{gt}))} \quad (50)$$

where $rel(v_{gt})$ is a binary function that returns 1 if the ground-truth item is in the top-K list, or else it will return 0.

We set K as 10 and 20.

5.3. Baselines

We compare MLEUP with the following four groups of baselines:

Traditional SBR method:

- **FPMC** [10]: It is a sequential method based on Markov chain.

Attention and RNN based SBR methods:

- **GRU4Rec** [17]: It is a model based on RNN that uses GRU to model sequence information.

² <https://tianchi.aliyun.com/dataset/dataDetail?dataId=42>

³ https://competitions.codalab.org/competitions/11161#learn_the_details-data2

⁴ <https://recsys.deib.polimi.it/datasets/>

- **NARM** [18]: It incorporates attention mechanisms into the RNN-based model to capture the user's main intention.
- **Atten-Mixer** [20]: It is a deparameterized and efficient model, removing the propagation part of GNN and retaining the readout module.

GNN-based SBR methods:

- **SR-GNN** [21]: It is a pioneering GNN-based method that uses GGNN to capture pairwise transition relationships between items within a session.
- **GCE-GNN** [5]: It is a GNN-based model that leverages both the global graph and session graph to model the transition patterns between items.
- **MGIR** [26]: It learns three types of global item relationships through session-independent and data-driven ways, using different graph construction methods.
- **Disen-GNN** [25]: It explores finer-granularity interests of users and generates session embedding at the factor level.

SSL/CL enhanced SBR methods:

- **S²-DHCN** [3]: It utilizes hypergraphs to capture high-order dependency among items and employs SSL to improve performance.
- **COTREC** [30]: It combines SSL with co-training to mine informative and practicable self-supervision signals.
- **PIE-CL** [32]: It proposes position-aware importance extraction module with CL to discover extra knowledge.
- **SPARE** [33]: It integrates a supervised CL task to tackle data sparsity and the inherent popularity bias.

5.4. Implementation details

Atten-Mixer,⁵ SR-GNN,⁶ GCE-GNN,⁷ MGIR,⁸ Disen-GNN,⁹ S²-DHCN,¹⁰ COTREC,¹¹ PIE-CL,¹² SPARE¹³ were provided by the authors. FPMC,¹⁴ GRU4Rec,¹⁵ NARM¹⁶ were implemented based on public resources. Following previous methods [8,21], we set the embedding size d of the baseline models and our proposed model to 100. In addition, for the FPMC baseline, we introduced the user information necessary for the model, and for the GRU4Rec baseline, we chose the widely used cross-entropy as the loss function. For each baseline, all other hyper-parameters were set following the suggestions from the original settings in their papers and we reported each baseline performance under its optimal settings.

Grid search was applied for our model to find the optimal hyper parameters using the last 10% of the training set as the validation set. For our proposed model, we set the batch size for mini-batch to 100, the L_2 regularization to 10^{-4} , the proportion of long-tail items δ to 0.8, the proportion of sampled items for each attribute value ω to 0.2, the maximum length of the sliding window z to 20, the range of high-order neighbors for items θ to 3, and the number of heads for multi-head self-attention h to 4. In addition, the strength λ of the self-supervised task and m for building a session group were selected from $\{0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$ and $\{10, 12, 14, 16\}$. For Tmall and

30music datasets, the threshold μ for achieving an incompatibility relationship was selected from $\{5, 10, 15, 20\}$. For Diginetica dataset, μ was selected from $\{15, 20, 25, 30\}$. All parameters were initialized using a Gaussian distribution with a mean of 0 and a standard deviation of 0.1, and the Adam optimizer was exerted to optimize these parameters, where the initial learning rate was set to 0.001 and would decay by 0.1 after every 3 epochs.

In Sections 5.5 and 5.6, the three hyper-parameters are set to relatively optimal values. For Tmall dataset, $\lambda = 0.3$, $\mu = 10$, $m = 14$; For Diginetica dataset, $\lambda = 0.05$, $\mu = 20$, $m = 12$; For 30music dataset, $\lambda = 0.2$, $\mu = 15$, $m = 12$.

5.5. Comparison with baselines

Table 3 shows the overall performance, from which we have the following key conclusions:

The performance of the conventional method is not competitive. FPMC relies on user profiles, is not suitable for session-based recommendation scenarios and cannot consider other important information such as the high-order sequential relationship between items.

Three attention and RNN based methods surpass the performance of the conventional method in all three datasets, indicating the advantages of incorporating sequential information in session-based recommendation. GRU4Rec without attention does not outperform NARM and Atten-Mixer, which suggests the effectiveness of the attention mechanism.

Four GNN-based methods in general perform better than the first two RNN-based methods, indicating that GNNs can capture complex transition relationships. Atten-Mixer almost outperforms SR-GNN and Disen-GNN, indicating the effectiveness of its proposed multi-level reasoning component in the attention-based readout module. Among the four, SR-GNN performs the worst because it only utilizes information within a single session. Disen-GNN outperforms SR-GNN because although it focuses on a single session, it explores more finer-granularity interests of users. GCE-GNN and MGIR achieve better performance by incorporating global information. MGIR performs better on the Tmall dataset, indicating that independently modeling different item relationship patterns and incorporating negative relationship information is more effective for the Tmall dataset. GCE-GNN models four types of first-order relationships and global co-occurrence relationships, achieving outstanding performance on the latter two datasets.

Three SSL/CL enhancement methods (SPARE, COTREC, and S²-DHCN) perform well on all datasets and basically outperform GNN-based methods. indicating the advantage of using self-supervised learning or contrastive learning. Among them, SPARE achieves the best performance. It finds positive samples for the current session through the modified session similarity calculation strategy and optimizes the negative sample sampling method. It shows the importance of finding accurate similar sessions and designing effective contrastive learning methods. PIE-CL performs poor, especially on the Tmall dataset, because it emphasizes the adjacency relationship between items by introducing a position-aware affinity-matrix, which fails to effectively decompose the mixed dependencies among items.

Our proposed MLEUP is better than almost all baselines, especially on the Tmall dataset, followed by diginetica dataset, with less improvement from 30music dataset. There are two possible reasons for this: The main reason is that Tmall dataset has the most available attribute categories, with a total of 3, while diginetica dataset has two, while 30music dataset has only one. More attribute knowledge provides enough data for MLEUP's attribute learning module, improving the quality of attribute embedding learning. It indirectly affects the quality of learning of both preferences. Second, the two relationships modeled by MLEUP are based on the constructed session groups, and the introduction of more sessions has the most obvious improvement on the Tmall dataset, while the reliable session groups we built provides valuable information to the training process. which is consistent with

⁵ <https://github.com/Peiyance/Atten-Mixer-torch>

⁶ <https://github.com/CRIPAC-DIG/SR-GNN>

⁷ <https://github.com/CCIPLab/GCE-GNN>

⁸ <https://github.com/zc-97/MGIR>

⁹ <https://github.com/AnsongLi/Disen-GNN>

¹⁰ <https://github.com/xiaxin1998/DHCN>

¹¹ <https://github.com/xiaxin1998/COTREC>

¹² <https://github.com/judiebig/PIE-CL>

¹³ <https://github.com/dbis-uibk/SPARE>

¹⁴ <https://github.com/khesui/FPMC>

¹⁵ <https://github.com/yeganegi-reza/Torch-GRU4Rec>

¹⁶ <https://github.com/Wang-Shuo/Neural-Attentive-Session-Based-Recommendation-PyTorch>

Table 3

Performances (%) of all comparison methods on three datasets. The best and second-best results are highlighted in boldface and underlined, respectively.

	Tmall						Diginetica						30music					
	P@10	M@10	N@10	P@20	M@20	N@20	P@10	M@10	N@10	P@20	M@20	N@20	P@10	M@10	N@10	P@20	M@20	N@20
FPMC	10.42	5.51	6.67	12.70	5.67	7.25	13.32	5.65	7.14	23.01	5.92	7.93	5.50	2.95	3.55	7.04	3.06	3.94
GRU4Rec	16.74	9.62	11.31	19.71	9.82	12.06	29.15	13.85	17.45	38.47	14.49	19.80	22.12	12.66	14.92	24.75	12.85	15.59
NARM	26.29	15.27	17.89	31.14	15.61	19.12	37.89	16.44	21.46	51.12	17.35	24.79	35.40	18.96	22.94	38.85	19.21	23.82
Atten-Mixer	28.04	15.93	18.80	33.36	16.30	20.15	40.44	17.62	22.97	53.93	18.55	26.37	39.22	23.00	26.95	42.49	23.24	<u>27.79</u>
SR-GNN	23.23	13.20	15.58	27.74	13.51	16.72	38.43	16.83	21.89	51.33	17.72	25.15	34.30	21.45	24.54	37.79	21.69	25.43
GCE-GNN	27.74	14.92	17.97	32.79	15.27	19.25	<u>41.03</u>	<u>18.13</u>	<u>23.51</u>	<u>54.13</u>	<u>19.04</u>	<u>26.82</u>	38.89	<u>23.15</u>	<u>26.94</u>	42.70	23.42	27.91
MGIR	30.67	17.06	20.29	36.16	17.44	21.38	40.71	17.90	23.26	53.76	18.81	26.56	39.21	17.58	22.75	43.58	17.90	23.87
Disen-GNN	24.79	14.40	16.87	29.06	14.70	17.95	40.42	18.01	23.27	53.20	18.90	26.50	36.88	<u>23.01</u>	26.34	40.40	23.26	27.23
S ² -DHCN	28.81	16.18	19.16	34.85	16.60	20.70	39.49	17.14	22.38	52.90	18.07	25.77	40.45	18.30	23.64	45.23	18.64	24.86
COTREC	30.26	17.48	20.50	36.35	17.90	22.05	40.41	17.61	22.96	54.00	18.56	26.40	<u>40.52</u>	18.85	24.11	44.78	19.17	25.22
PIE-CL	19.80	10.76	12.90	23.79	11.03	13.91	38.25	16.27	21.41	51.50	17.19	24.76	35.84	23.15	26.21	39.24	<u>23.39</u>	27.07
SPARE	<u>33.55</u>	<u>19.36</u>	<u>22.73</u>	<u>39.39</u>	<u>19.76</u>	<u>24.21</u>	40.41	17.64	22.97	53.59	18.55	26.30	40.16	15.48	21.33	<u>45.51</u>	15.86	22.69
MLEUP	41.23	20.37	25.31	49.65	20.96	27.45	42.24	18.52	24.09	55.39	19.43	27.41	40.66	19.05	24.23	46.18	19.44	25.64
Improv.	22.89%	5.22%	11.35%	26.05%	6.07%	13.38%	2.95%	2.15%	2.47%	2.33%	2.05%	2.20%	0.35%	–	–	1.47%	–	–

Table 4

Impact of modeling different item relationships on recommendation performance.

	Tmall				Diginetica				30music			
	P@10	M@10	P@20	M@20	P@10	M@10	P@20	M@20	P@10	M@10	P@20	M@20
MLEUP (w/o seq)	39.95	19.87	48.51	20.46	42.08	18.42	55.30	19.34	40.68	18.63	46.15	19.02
MLEUP (w/o co)	40.74	20.89	49.09	21.47	40.10	17.59	53.15	18.50	40.60	19.40	46.17	19.80
MLEUP	41.23	20.37	49.65	20.96	42.24	18.52	55.39	19.43	40.66	19.05	46.18	19.44

the experimental results shown in Table 5. Another observation is that on the 30music dataset, most baseline models and MLEUP failed to improve both precision and ranking (MRR and NDCG). This may be due to the fact that the 30music dataset reflects users' listening preferences for music. Compared with the datasets of the first two e-commerce scenarios, users are more likely to repeat the same song they have heard and loved before in a session, or they may also set up loop playback, so there are more repeats in a session. Therefore, models (SR-GNN, Disen-GNN and GCE-GNN) that focuses more on modeling sequential relationships within a session can collect fewer items that are similar to the target item, and it is not easy to mix in more similar items in the ranking, which reduces the probability of predicting the target item, but it is easy to put the target item in the front. In contrast, models (MGIR, S²-DHCN, COTREC and SPARE) that learn more information from inter-session are more likely to predict target item, and also pay attention to more items that are similar to the target item, thereby lowering the ranking of the target item. However, through the learning process of multi-granularity co-occurrence relationship and high-order sequential relationship of MLEUP, it can make better use of inter-session collaborative information, and can achieve the optimal precision, and it also performs well on MRR and NDGC, achieving good performance in balance.

5.6. Ablation study

In this section, we design three ablation experiments to investigate different questions: (1) The contribution of modeling different item relationships to the improvement of MLEUP's performance. (2) Whether the session group constructed by MLEUP introduces more useful information to the training process. (3) Whether MLEUP learns valid attribute information. We use precision and MRR as evaluation metrics.

5.6.1. Impact of item relationships

To verify the contributions of modeling different item relationships in MLEUP, we conduct an ablation study with two variants:

- MLEUP (w/o seq): MLEUP without modeling item sequential relationship.

- MLEUP (w/o co): MLEUP without modeling item co-occurrence relationship.

Table 4 shows the performances of different variants on the three datasets. It can be observed that each item relationship contributes to performance. For Tmall dataset, the performance of MLEUP (w/o co) is better than that of MLEUP (w/o seq), indicating that modeling high-order sequential relationship is more important for improving recommendation performance. For Diginetica dataset, the opposite conclusion is reached. The variant of MLEUP (w/o seq) that only models multi-granularity co-occurrence relationship performs almost as well as the MLEUP, indicating that this dataset is more affected by co-occurrence relationship. Our proposed model MLEUP encodes both high-order sequential relationship and multi-granularity co-occurrence relationship, as well as balances the two with a gating mechanism, ensuring that useful information is learned from two types of relationship without introducing noise, achieving optimal performance. For 30music dataset, in terms of precision, the variant of MLEUP (w/o seq) performs roughly on par with the variant of MLEUP (w/o co) but slightly lower than MLEUP. However, in terms of MRR, the variant of MLEUP (w/o co) outperforms the others, followed by MLEUP, while the variant of MLEUP (w/o seq) performs the worst. This indicates that in the music dataset, sequential relationships are more important compared to co-occurrence relationships. Modeling sequential relationships with position coding can help improve the ranking of target items. Our proposed model, MLEUP, achieves a satisfactory performance by ensuring a leading precision while also demonstrating a superior performance in MRR.

5.6.2. Impact of session group

To demonstrate the effectiveness of our proposed session group strategy in MLEUP, we conduct an ablation study with two variants:

- MLEUP (w/o sg): MLEUP without constructing a session group. Therefore, the learning of the two item relationships is solely based on the session itself, and the self-supervised task is removed.
- MLEUP (simple sg): We use a simple cosine similarity as the measure of session similarity, while keeping the other parts the same as MLEUP.

Table 5
Impact of different session group strategies on recommendation performance.

	Tmall				Diginetica				30music			
	P@10	M@10	P@20	M@20	P@10	M@10	P@20	M@20	P@10	M@10	P@20	M@20
MLEUP (w/o sg)	27.81	14.75	34.05	15.18	38.48	16.56	51.40	17.45	37.02	20.85	42.88	21.26
MLEUP (simple sg)	40.13	19.36	48.53	19.95	41.96	18.35	55.07	19.26	40.44	18.68	45.98	19.07
MLEUP	41.23	20.37	49.65	20.96	42.24	18.52	55.39	19.43	40.66	19.05	46.18	19.44

Table 6
Impact of different attribute learning strategies on recommendation performance.

	Tmall				Diginetica				30music			
	P@10	M@10	P@20	M@20	P@10	M@10	P@20	M@20	P@10	M@10	P@20	M@20
MLEUP (w/o ssl)	30.76	15.72	38.19	16.24	40.68	17.59	54.08	18.52	36.92	20.72	42.80	21.13
MLEUP (w/o att)	32.49	16.27	39.79	16.76	40.83	17.63	53.73	18.53	35.61	20.77	39.92	21.08
MLEUP	41.23	20.37	49.65	20.96	42.24	18.52	55.39	19.43	40.66	19.05	46.18	19.44

Table 5 shows the performances of different variants on the three datasets. MLEUP outperforms the variant MLEUP (simple sg) on all three datasets, indicating that our proposed method of using positive and negative relationships between items to calculate session similarity can better capture the complex relationships between sessions and build more reliable session groups, thus avoiding the introduction of excessive noise. Additionally, for the first two datasets, MLEUP outperforms the variant MLEUP (w/o sg), indicating that introducing global information through the construction of session groups can significantly improve recommendation performance. For 30music dataset, the results differ. MLEUP is significantly better than variant MLEUP (w/o sg) in precision, but poor in MRR, which may be due to the fact that after learning more collaborative information, it is easier to predict more target items while also paying attention to more items similar to the target items, thus lowering the ranking of the target items to some extent. But MLEUP's greater advantage in precision still makes it more competitive.

5.6.3. Impact of attribute learning

To verify whether we have learned useful attribute information in MLEUP, we conduct an ablation study with two variants:

- MLEUP (w/o ssl): MLEUP without neighbor-based contrastive self-supervised attribute learning task.
- MLEUP (w/o att): MLEUP without using attribute knowledge. Therefore, we only use the dynamic preference generated by the item relationships learning as the final preference of the user.

Table 6 shows the performances of different variants on the three datasets. For Tmall dataset, it can be observed that the performance of MLEUP is significantly better than that of the variant MLEUP (w/o att), indicating that incorporating attribute information can help improve recommendation performance. The same trend is also observed for Diginetica dataset, but this dataset is less affected by attribute information. Additionally, the performance of MLEUP is better than the variant MLEUP (w/o ssl) on three datasets, indicating that if attribute information is not properly modeled, the performance of the recommendation may even decrease. For 30music dataset, MLEUP is significantly better in precision than the two variants, but worse in MRR, suggesting that the additional knowledge introduced by attributes, as well as the collaboration information introduced by session groups, provided more information, but indirectly led to a decline in the ranking of the target items. On the whole, our designed neighbor-based contrastive self-supervised attribute learning task can obtain informative supervised signals from the correlation between sessions, guide the learning of attribute embeddings, which has a positive effect on the performance of the recommendation task.

5.7. Impact of hyper-parameters

We further study the impact of the key hyper-parameters m , λ and μ , and report the results on precision and MRR metrics in Fig. 4.

The hyper-parameter λ controls the strength of self-supervised learning, and we report the performance of MLEUP with a set of representative λ values {0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5}. For Tmall dataset, the performance of MLEUP improves rapidly with increasing λ until reaching the maximum values for P@20 and MRR@20 at $\lambda = 0.3$, after which it starts to decline. For Diginetica dataset, the performance of MLEUP reaches its maximum at a smaller value of $\lambda = 0.05$, and then declines but remains relatively stable. This may be due to the fact that the introduction of more attribute information has a larger impact on the training of Tmall dataset. When the value of λ is too small, the self-supervised signals provided may not be sufficient to improve the quality of attribute embedding learning. Conversely, when the value of λ is too large, it introduces too much self-supervised signal, leading to overfitting. On the other hand, Diginetica dataset is less sensitive to the introduction of attribute knowledge, hence requiring fewer self-supervised signals to improve recommendation performance. For 30music dataset, as the value of λ increases, P@20 rapidly increases and then levels off, and MRR@20 rapidly decreases and then slowly declines, indicating that with the introduction of more attribute knowledge, it becomes easier to predict target items while also considering more items similar to the target items, which leads to a decrease in the ranking of the target items.

The hyper-parameter μ controls the threshold for reaching an incompatibility relationship. Due to the lower item frequency in Tmall and 30music datasets, μ is selected from {5, 10, 15, 20} for the two datasets, and for Diginetica dataset, μ is selected from {15, 20, 25, 30}. For Tmall dataset, the performance of MLEUP shows an overall trend of first improvement and then decline with the increase of μ . The optimal performance is achieved at $\mu = 10$. For Diginetica and 30music datasets, a similar pattern is observed, but with smaller fluctuations, and the optimal performance of MLEUP is achieved at $\mu = 20$ and $\mu = 15$, respectively. This suggests that when μ is larger, fewer incompatibility relationships are captured in the statistics, which may not effectively contribute to the construction of session groups. On the other hand, when μ is smaller, more unreliable incompatibility relationships may be introduced, which could impair the performance of recommendation. Therefore, selecting an appropriate threshold value for μ can help improve the performance of the model.

The hyper-parameter m denotes the number of sessions in the session group, and we report the performance of MLEUP with a set of m values {10, 12, 14, 16}. For Tmall and Diginetica datasets, the performance of MLEUP shows a trend of first increase and then decrease with the increase of m , but the overall performance is relatively stable. The optimal performance of MLEUP is achieved at $m = 14$ and $m = 12$, respectively. This may be because a smaller m does not provide enough collaboration information, while a larger m introduces only a small

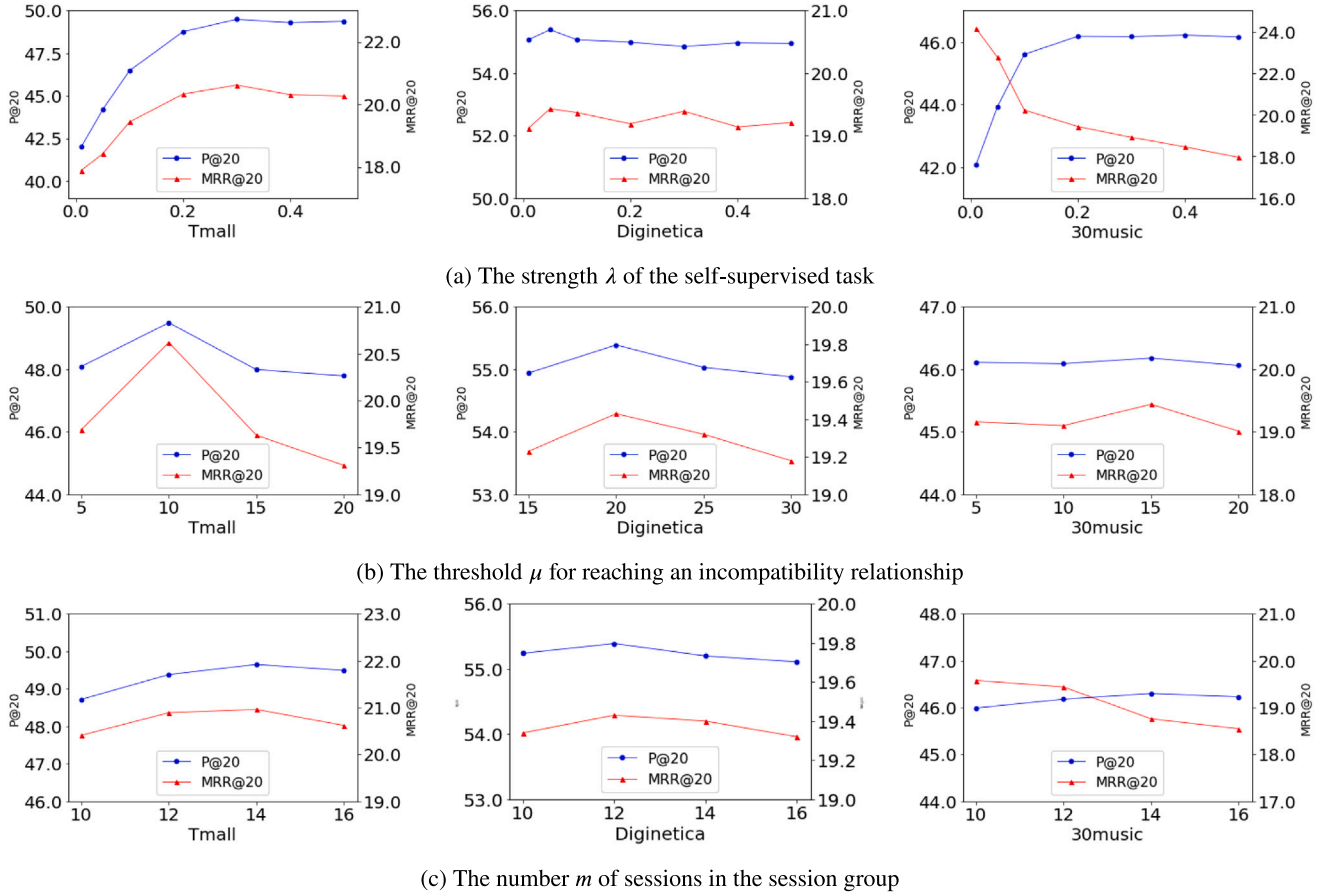


Fig. 4. Impact of different hyper-parameters.

amount of noise data due to the reliable session group we built. For 30music dataset, as the value of m increases, P@20 slowly increases, and MRR@20 slowly decreases, this may have the same reason as introducing more attribute information.

6. Conclusion

In this paper, we propose the MLEUP, which captures user's stable preference through modeling item attribute knowledge and captures user's dynamic preference through comprehensive modeling of multi-granularity item co-occurrence relationship and high-order item sequential relationship. By incorporating attribute knowledge and global context, we resolve the data sparsity problem, and by comprehensively modeling item relationships, we capture complex user preferences. Finally, extensive experiments on three benchmark datasets demonstrate the effectiveness of MLEUP in terms of precision, MRR and NDCG.

However, through experiments, we have found that considering attribute information increases the time required for model training. This could be attributed to the large number of sample items associated with each attribute value. For our future work, we plan to improve our method from the following perspectives. First, we will optimize our attribute learning scheme by adopting more refined sampling methods, such as narrowing down the samples to the constructed session groups. Second, by leveraging the constructed session groups, we aim to minimize the noise during the attribute training process, thereby further enhancing the quality of attribute embedding learning.

CRedit authorship contribution statement

Lan Zhang: Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Derong Shen:** Writing – original draft,

Validation, Methodology, Conceptualization. **Yue Kou:** Writing – original draft, Validation, Methodology, Conceptualization. **Tiezheng Nie:** Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study are publicly available.

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